

Apache JMeter Performance Testing Report Case Study 2

1. Objective

The objective of this performance testing activity was to evaluate the response time, reliability, and overall performance of multiple web pages using Apache JMeter.

2. Tool Used

- Apache JMeter 5.6.3
- Java JDK
- Windows Operating System

3. Test Environment

- Tool Version: Apache JMeter 5.6.3
- Protocol: HTTP/HTTPS
- Request Type: HTTP Request
- Test Execution Mode: GUI and Non-GUI
- Report Type: HTML Dashboard Report

4. Test Scenarios Executed

The following modules were tested using separate Thread Groups:

4.1 Calculations Module

- HTTP Request executed successfully
- Response Code: 200 OK
- Total Samples: 800
- Average Response Time: 99 ms
- Status: Passed

4.2 Convertor Module

- HTTP Request executed successfully
- Response Code: 200 OK
- Average Response Time: 253 ms
- Status: Passed

4.3 Formulas Module

- HTTP Request executed successfully
- Response Code: 200 OK
- Average Response Time: 255 ms
- Status: Passed

4.4 Currencies Module

- HTTP Request executed successfully
- Response Code: 200 OK
- Average Response Time: 316 ms
- Status: Passed

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4.5 Charts Module

- HTTP Request executed successfully
- Response Code: 200 OK
- Total Samples: 50
- Average Response Time: 452 ms
- Status: Passed

4.6 Example Module

- HTTP Request executed successfully
- Response Code: 200 OK
- Average Response Time: 263 ms
- Status: Passed

4.7 Tutorials Module

- HTTP Request executed successfully
- Response Code: 200 OK
- Average Response Time: 256 ms
- Status: Passed

4.8 Answers Module

- HTTP Request executed successfully
- Response Code: 200 OK
- Total Samples: 50
- Average Response Time: 561 ms
- Status: Passed

5. Execution Results

Sample Result Verification

The following validations were performed:

- Response Code = 200
- Response Message = OK
- Error Count = 0
- Successful Request Execution
- Proper Data Encoding (UTF-8)

All requests executed successfully without failures.

6. HTML Dashboard Analysis

Test and Report Information

- Source File: result.jtl
- Dashboard Generated Successfully
- Report Generated Using Apache JMeter Dashboard Generator

Link : <file:///D:/JMeterTests/HTMLReport/index.html>

APDEX (Application Performance Index)

- APDEX Score: 0.545
- Interpretation:
- Application performance is acceptable.
- Majority of users received responses within the acceptable threshold.
- No critical performance degradation observed.

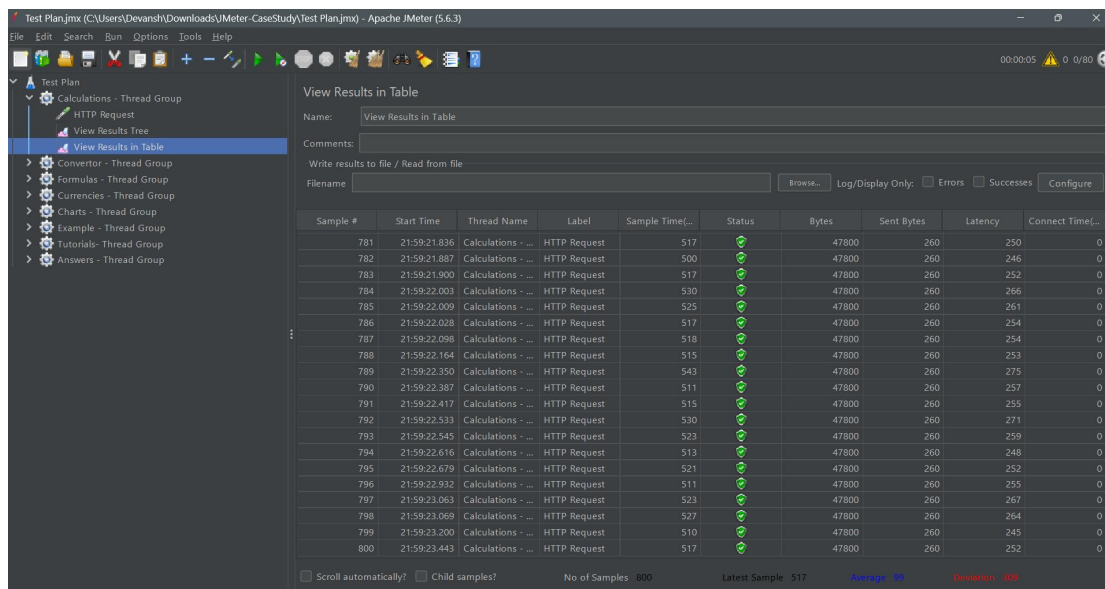
Request Summary

- PASS: 100%
- FAIL: 0%
- Observation:
- All requests passed successfully.
- No failed transactions were recorded during execution.

7. Testing Links:

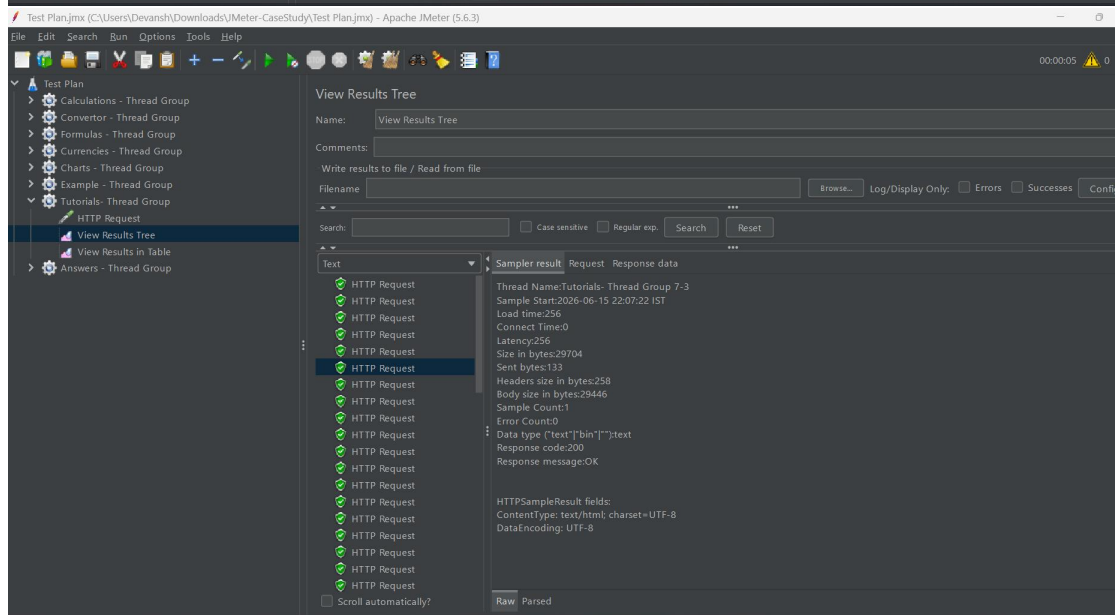
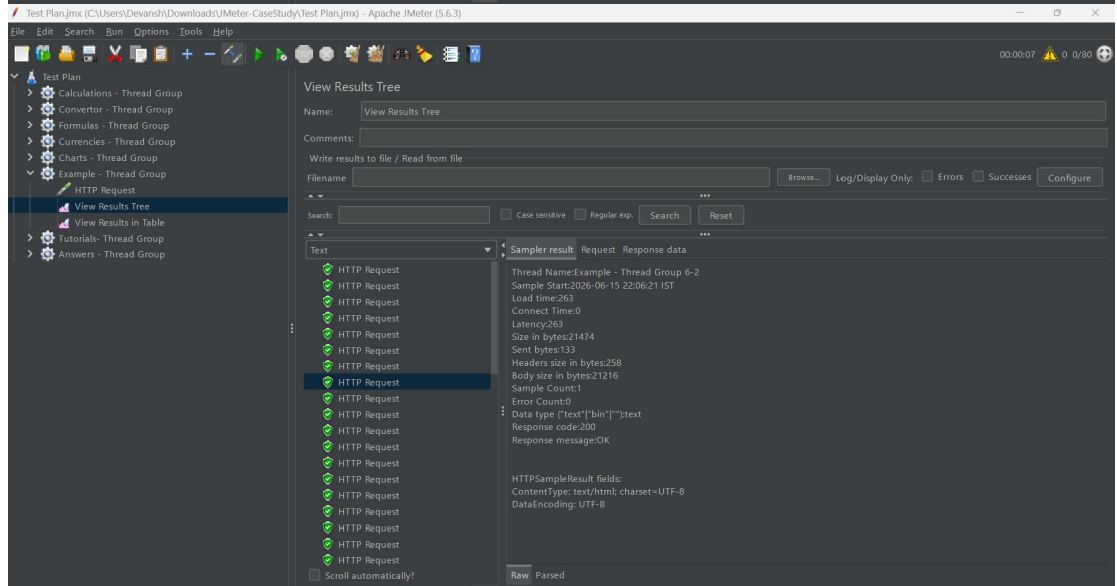
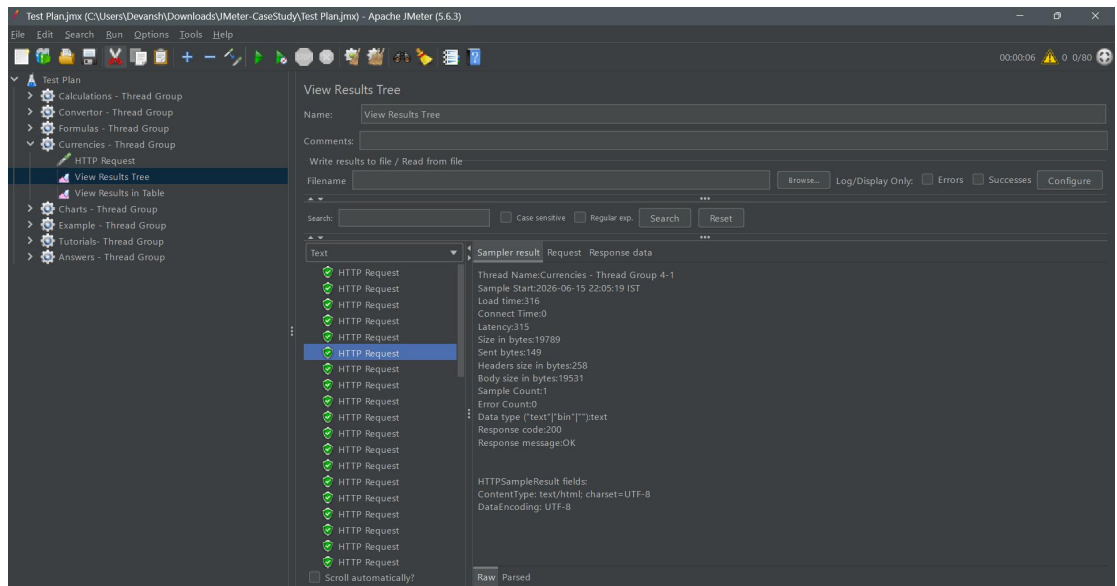
- Link1: <https://www.easycalculation.com/>
- Link2: <https://www.easycalculation.com/unit-conversion/index.php>
- Link3: <https://www.easycalculation.com/formulas/index.php>
- Link4: <https://www.easycalculation.com/currency-converter/index.php>
- Link5: <https://www.easycalculation.com/charts-and-tables.php>
- Link6: <https://www.easycalculation.com/examples.php>
- Link7: <https://www.easycalculation.com/tutorial.php>
- Link8: <https://www.easycalculation.com/faq/index.php>

8. ScreenShots



The screenshot shows the Apache JMeter Test Plan interface. The 'View Results in Table' window is open, displaying a table of test results. The table has columns for Sample #, Start Time, Thread Name, Label, Sample Time, Status, Bytes, Sent Bytes, Latency, and Connect Time. The data shows 800 samples, all with a status of 'Success' (indicated by green checkmarks). The test plan includes a 'Test Plan' node with a 'Thread Group' containing 'Calculations - Thread Group', which is further divided into 'HTTP Request', 'Convertor - Thread Group', 'Formulas - Thread Group', 'Currencies - Thread Group', 'Charts - Thread Group', 'Example - Thread Group', 'Tutorials - Thread Group', and 'Answers - Thread Group'.

Sample #	Start Time	Thread Name	Label	Sample Time	Status	Bytes	Sent Bytes	Latency	Connect Time
701	21:59:21.835	Calculations - ...	HTTP Request	517	Success	47800	260	250	0
702	21:59:21.887	Calculations - ...	HTTP Request	500	Success	47800	260	246	0
703	21:59:21.900	Calculations - ...	HTTP Request	517	Success	47800	260	252	0
704	21:59:22.003	Calculations - ...	HTTP Request	530	Success	47800	260	266	0
705	21:59:22.009	Calculations - ...	HTTP Request	525	Success	47800	260	261	0
706	21:59:22.028	Calculations - ...	HTTP Request	517	Success	47800	260	254	0
707	21:59:22.098	Calculations - ...	HTTP Request	518	Success	47800	260	254	0
708	21:59:22.164	Calculations - ...	HTTP Request	515	Success	47800	260	253	0
709	21:59:22.350	Calculations - ...	HTTP Request	543	Success	47800	260	275	0
710	21:59:22.387	Calculations - ...	HTTP Request	511	Success	47800	260	257	0
711	21:59:22.417	Calculations - ...	HTTP Request	515	Success	47800	260	255	0
712	21:59:22.533	Calculations - ...	HTTP Request	530	Success	47800	260	271	0
713	21:59:22.545	Calculations - ...	HTTP Request	523	Success	47800	260	259	0
714	21:59:22.616	Calculations - ...	HTTP Request	513	Success	47800	260	248	0
715	21:59:22.679	Calculations - ...	HTTP Request	521	Success	47800	260	252	0
716	21:59:22.932	Calculations - ...	HTTP Request	511	Success	47800	260	255	0
717	21:59:23.063	Calculations - ...	HTTP Request	523	Success	47800	260	267	0
718	21:59:23.069	Calculations - ...	HTTP Request	527	Success	47800	260	264	0
719	21:59:23.200	Calculations - ...	HTTP Request	510	Success	47800	260	245	0
720	21:59:23.443	Calculations - ...	HTTP Request	517	Success	47800	260	252	0



Test Plan.jmx (C:\Users\Devansh\Downloads\JMeter-CaseStudy\JMeter-Test Plan.jmx) - Apache JMeter (5.6.3)

File Edit Search Run Options Tools Help

Test Plan

- Calculations - Thread Group
- Converter - Thread Group
- Formulas - Thread Group
- Currencies - Thread Group
- Charts - Thread Group
- Example - Thread Group
- Tutorials - Thread Group
- Answers - Thread Group
 - HTTP Request
 - View Results Tree
 - View Results in Table

View Results in Table

Name: View Results in Table

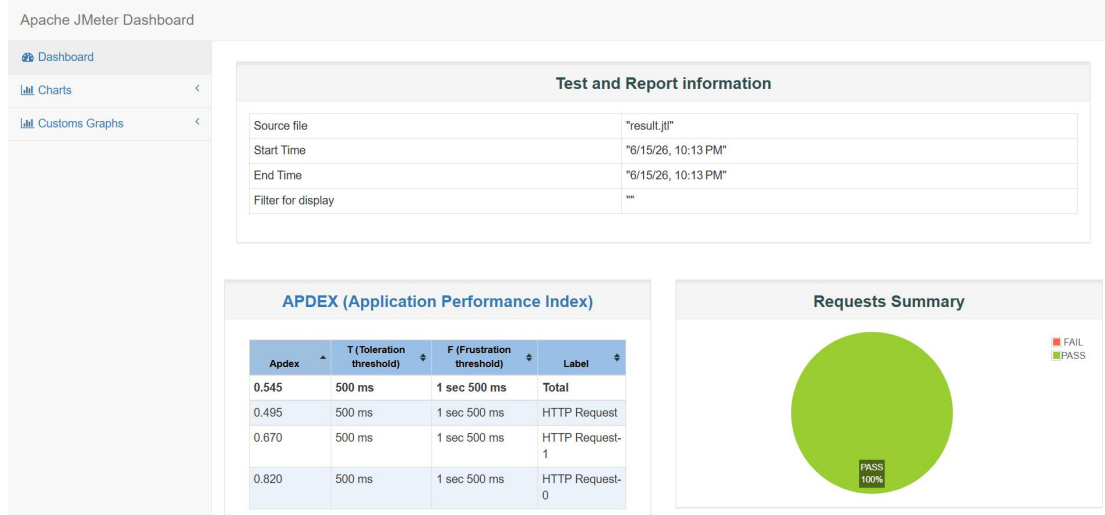
Comments:

Write results to file / Read from file

Filename: Browse... Log/Display Only: ☐ Errors ☐ Successes ☐ Configure

Sample #	Start Time	Thread Name	Label	Sample Time...	Status	Bytes	Sent Bytes	Latency	Connect Time...
1	22:08:16.254	Answers - Thre...	HTTP Request	1125	✓	59765	134	817	541
2	22:08:15.653	Answers - Thre...	HTTP Request	1855	✓	59765	134	1599	1095
3	22:08:15.854	Answers - Thre...	HTTP Request	1724	✓	59765	134	1203	915
4	22:08:16.054	Answers - Thre...	HTTP Request	1546	✓	59765	134	1263	715
5	22:08:17.379	Answers - Thre...	HTTP Request	297	✓	59765	134	288	0
6	22:08:16.449	Answers - Thre...	HTTP Request	1331	✓	59765	134	1052	527
7	22:08:17.508	Answers - Thre...	HTTP Request	297	✓	59765	134	297	0
8	22:08:17.578	Answers - Thre...	HTTP Request	322	✓	59765	134	322	0
9	22:08:17.600	Answers - Thre...	HTTP Request	338	✓	59765	134	310	0
10	22:08:17.676	Answers - Thre...	HTTP Request	293	✓	59765	134	293	0
11	22:08:16.653	Answers - Thre...	HTTP Request	1339	✓	59765	134	1050	500
12	22:08:17.780	Answers - Thre...	HTTP Request	321	✓	59765	134	317	0
13	22:08:16.852	Answers - Thre...	HTTP Request	1250	✓	59765	134	897	565
14	22:08:17.805	Answers - Thre...	HTTP Request	323	✓	59765	134	323	0
15	22:08:17.901	Answers - Thre...	HTTP Request	288	✓	59765	134	288	0
16	22:08:17.938	Answers - Thre...	HTTP Request	291	✓	59765	134	275	0
17	22:08:17.969	Answers - Thre...	HTTP Request	315	✓	59765	134	295	0
18	22:08:17.992	Answers - Thre...	HTTP Request	319	✓	59765	134	318	0
19	22:08:18.102	Answers - Thre...	HTTP Request	324	✓	59765	134	324	0
20	22:08:18.101	Answers - Thre...	HTTP Request	333	✓	59765	134	333	0

☐ Scroll automatically? ☐ Child samples? No of Samples: 50 Latest Sample: 504 Average: 561 Deviation: 476



9. Performance Metrics

Metric	Value
Total Requests	800+
Successful Requests	100%
Failed Requests	0%
APDEX Score	0.545
Response Code	200 OK
Error Count	0
Maximum Average Response Time	561 ms
Minimum Average Response Time	99 ms

10. Observations

1. All HTTP Requests returned a 200 OK status.
2. No request failures were observed.
3. The application handled the configured load successfully.
4. Response times remained within acceptable limits.
5. The APDEX score indicates satisfactory user experience.
6. The HTML dashboard confirms 100% successful request execution.

11. Conclusion

The performance testing was successfully completed using Apache JMeter 5.6.3. All modules including Calculations, Convertor, Formulas, Currencies, Charts, Example, Tutorials, and Answers executed successfully with a 100% pass rate. No failures or errors were encountered during testing. The application demonstrated stable performance under the configured load, and the generated HTML dashboard confirmed the reliability and responsiveness of the tested application.

Result: PASSED

Overall Application Status: STABLE